

Examiner reports

Q1.

A significant number of students correctly stated that $\tan 15 = \frac{\sin 15}{\cos 15}$ but, disappointingly, then wrote down the given surd forms of $\cos 15$ divided by $\sin 15$. Many resorted to using their calculator, which gave the final answer immediately, but this approach received no credit as students were clearly asked to fully justify their answer. Some students did not appreciate the necessity of using the conjugate to rationalise.

Q2.

Many students made good progress with this question and chose an appropriate technique to be able to determine the transformations. Most identified $R \sin(x - \alpha)$ as the best approach, although some students chose $R \sin(x + \alpha)$ and a few individuals started with $R \sin(x \pm \alpha)$, which often resulted in sign errors.

Having decided on an approach, most students handled well the routine of finding R and α although some used some inefficient methods and did not seem sufficiently well practiced in what is a very standard technique.

In order to fully justify their answer a clear comparison of the given equation and their $R \sin(x \pm \alpha) + 4$ needed to be made and in a few cases students' reasoning was not clear. Most students achieved some credit for correctly describing transformations relating to their $R \sin(x \pm \alpha) + 4$. All was not lost if mistakes were made early in the solution.

Part (b)(i) hinged on students realising that the denominator had to be maximised and so their $\sin(x \pm \alpha)$ had to equal 1. Students who had gone wrong in part (a) could still achieve some credit in part (b)(i). Many students gave several lines of working to rationalise the denominator of their fraction. This was unnecessary given allowed calculator technology.

The mark for part (b)(ii) was available to students who had gone wrong in part (a). Provided they used their R correctly, work was followed through.

Q6.

- (a) (i) Most students were able to express $\sqrt{18}$ as $3\sqrt{2}$, although a few weaker students were unable to do this correctly. Some completely ignored this part of the question.
- (ii) The majority followed the expected method by writing each term as a multiple of $\sqrt{2}$ and most students then simplified their expression to $\frac{2\sqrt{2}}{7\sqrt{2}}$. A surprisingly large number of students left this as their final answer or presented the final answer as $\frac{2\sqrt{2}}{7}$. Another common approach was to

multiply by $\frac{\sqrt{18} - \sqrt{32}}{\sqrt{18} - \sqrt{32}}$, but poor arithmetic skills meant that many were unable to simplify $\frac{12 - 16}{18 - 32}$ correctly.

- (b) Many fully correct answers were seen to this familiar rationalising of the denominator request. The most common error was to not simplify $-2\sqrt{6} + 7\sqrt{6}$ correctly in the numerator, with the result often being given as $-5\sqrt{6}$. Some were unable to cope with the product $\sqrt{2} \times \sqrt{3}$, particularly in the denominator, where this was frequently seen as $\sqrt{5}$.

Q7.

- (a) Many students began by trying to complete the square for both the x terms and the y terms. These students, in an attempt to put $x = 0$, often simply wrote $(y - 2)^2 = 12$ and scored no marks. A surprisingly small number of students substituted $x = 0$ directly into the original equation; those who followed this method sometimes stopped at $y^2 - 4y = 12$, apparently being unable to solve the quadratic equation; others concluded that “ $y = 12$ or $y - 4 = 12$ ”, which is a bad error and not really expected at this level of mathematics. Those trying to solve $(y - 2)^2 = 16$ tended to miss the negative square root of 16.

Students who found that the radius was 5 sometimes used a diagram with a right angled triangle, two of the sides being 3 and 5. This enabled them to find the y -intercepts using the y -coordinate of C , by writing $y = 2 \pm 4$.

- (b) In their working for the previous part, many had already established the equation of the circle in the form $(x + 3)^2 + (y - 2)^2 = 25$ and could write down the value of the radius. However, the correct method was in doubt when 25^2 , or $\sqrt{25}$, instead of 25, appeared on the right hand side of the student's equation, even though in each of these cases the radius was then stated as being 5.

A significant number of students, despite the coordinates of the centre being given in the question, wrote down an incorrect circle equation such as $(x - 3)^2 + (y + 2)^2 = 25$ and once again concluded that the radius was equal to 5. Students cannot expect to score full marks for any answer that follows incorrect working.

- (c) (i) Although most students recognised that they should use the distance formula, many added 25 and 9 to make 36 not 34. There were also many examples of poor notation, such as $5^2 + 3^2 = \sqrt{34}$. There were many arithmetic slips due to the negative coordinate of C . Those who used an incorrect formula for the length of CP scored no marks at all.
- (ii) Some realised that $(2, 2)$ was one of the points of contact and then found PQ very easily. Failing to recognise the position of the right angle often led to many students writing $\sqrt{34 + 25}$ for the length of PQ . Many, who perhaps believed that they were using Pythagoras correctly, wrote $\sqrt{34} - \sqrt{25} = \sqrt{9}$ and this scored no marks, even though students obtained the correct value for the length of PQ . The mention of “tangent” often triggered attempts to find

equations rather than using the geometrical properties of the circle.

Q8.

- (a) (i) Most candidates were able to express $\sqrt{48}$ as $4\sqrt{3}$ although a minority then went on to give an answer of $2\sqrt{3}$.
- (ii) Candidates who began by writing $\sqrt{12} \times = 2\sqrt{3} \times$ were often successful as they equated this to $7\sqrt{3} - 4\sqrt{3}$ and obtained the equation $2\sqrt{3} \times = 3\sqrt{3}$. However some then subtracted $2\sqrt{3}$ from $3\sqrt{3}$ instead of dividing. A few left their answer as $\frac{3\sqrt{3}}{2\sqrt{3}}$ or “simplified” this to $\sqrt{3}$ either using poor cancellation or subtraction. A few candidates adopted a different approach. They multiplied each side of the equation by $\sqrt{12}$ or $\sqrt{3}$ to obtain an equation in integers. Another alternative approach was to square each side of the equation leading to $12x^2 = 27$. Those who used this method often gave their answer as $\pm\sqrt{\frac{3}{2}}$, failing to recognise that their squaring had introduced an extra incorrect solution.
- (b) The majority of candidates rationalised the denominator by multiplying the numerator and denominator by $2\sqrt{3} - \sqrt{5}$. It was pleasing to see many correct solutions as candidates were able to cope with the product of $\sqrt{3}$ and $\sqrt{5}$, possibly aided by the form of the printed answer. Once more there were some arithmetic errors; for example, $66 - 10 = 55$ was quite common. The given form of the answer, stating that m was an integer, should have alerted them to an error in their working.

Q9.

The majority of candidates failed to answer this question well.

- (a) (i) Many candidates were unsuccessful in completing the square as they did not realise the need to take out a factor of 2 first, so $2(x + 3)^2 - 4$ was a common incorrect answer. Many who made the correct approach did not handle the factor of 2 correctly and the expression $2\left(x + \frac{3}{2}\right)^2 + \frac{1}{4}$ was seen quite often. Others made sign errors so another common incorrect answer was $2\left(x + \frac{3}{2}\right)^2 + \frac{11}{4}$.
- (ii) Again there were few correct answers. Many candidates gave the value of x for which the expression was a minimum, namely $x = -\frac{3}{2}$. Those who had completed the square correctly often offered a pair of coordinates, namely $\left(-\frac{3}{2}, \frac{1}{2}\right)$, as their answer instead of the minimum value of $\frac{1}{2}$.

- (b) (i) Many candidates had no idea how to find the distance between the two points and attempts such as the sum of the products of both x coordinates and both y coordinates were seen. This may have been due to a lack of understanding of AB^2 . Some also interpreted it as $A \times B^2$. Those who began by simplifying $(3x + 9 - 5)^2$, rather than producing an expression with up to nine terms, usually went on to complete the proof successfully. A few made sign errors when subtracting the y coordinates, which led to errors. There was some poor squaring such as $(x + 3)^2 = x^2 + 9$, though this did not occur often. It was good to see that almost all included " $AB^2 =$ " in their solution.
- (ii) There were very few correct solutions here. Hardly any candidates heeded the instruction of the question to use 5 times their value from part (a)(ii); those who attempted this part often substituted a value for x (usually in the form of a fraction) into $5(2x^2 + 6x + 5)$ and the simplification defeated them. Even some of the most able candidates were unable to change $\sqrt{\frac{5}{2}}$ into the form $\frac{1}{2}\sqrt{10}$.

Q10.

In part (a) (i), many candidates misinterpreted $(3\sqrt{2})^2$ as $(3 + \sqrt{2})^2$ and these candidates often used a grid method to obtain $11 + 6\sqrt{2}$. Others gave the value as 36, but the correct answer was seen quite often.

In part (a)(ii), a few weak candidates ignored the terms in $\sqrt{2}$ and wrote $18 + 1 + 9 + 2 = 30$, which earned no marks. This is a warning to candidates that getting the correct answer by an incorrect method will not be given any credit. Multiplication of surds caused some problems: $(\sqrt{2})^2$ was often seen evaluated as 4 instead of 2, and the arithmetic error $1 \times 1 = 2$ was seen quite a few times. Nevertheless, there were several correct solutions.

In part (b), once again, poor arithmetic led to errors such as $20 - 2 = 16$ and $40 + 14 = 56$. Most recognised the first crucial step of multiplying the numerator and denominator by

$2\sqrt{5} - \sqrt{2}$ and many obtained $\frac{54 - 18\sqrt{5}\sqrt{2}}{18}$, but then incorrect evaluation of the numerator or poor cancellation led to many failing to obtain the correct final answer. Some were unable to cope with the product of $\sqrt{5}$ and $\sqrt{2}$, which became $\sqrt{7}$ for example.

Q11.

Most candidates had part (a)(i) correct, although the amount of work they did varied considerably. Some just wrote down the value of $\tan \alpha$, presumably recognising the 3-4-5 triangle, whereas others used Pythagoras. Several weaker candidates divided $\sin \beta$ by $\cos \alpha$ to give an answer of $\frac{5}{6}$. Some candidates used trigonometric identities, some finding $\sin \alpha$ first and using it together with the given cosine to calculate the tangent.

In part (a)(ii), the majority of candidates failed to work with a negative value for $\tan \beta$, despite most of them showing awareness of an obtuse angle through quoting 150° . These candidates then could not get to the right answer in part (b).

However, in part (b), most candidates did use the compound angle formula for $\tan(\alpha + \beta)$ correctly, with only a few sign errors seen and some leaving the $\pm$, from the formula book, in their expression. Most tried to manipulate the surds in their expression towards the required form of the answer, but with varying degrees of success. Those who inverted the denominator and then multiplied the resulting fractions were the more successful. Some who had worked with the correct value of $\tan \beta$ failed to get to the required form by not seeing that they could cancel or multiply through their expression by a factor of $\sqrt{3}$.

Q12.

This surd question involving the rationalising of the denominator proved to be a good opener to the paper with many students offering a complete solution. Some used a grid method to evaluate the numerator and denominator, but with mixed success. The most common error occurred when multiplying $5\sqrt{3}$ by $2\sqrt{3}$ in the numerator. A few students

who obtained $\frac{48 - 27\sqrt{3}}{3}$ wrote their final answer as $16 - 27\sqrt{3}$.

Q13.

Part (a) was answered surprisingly badly, even though in some cases $(3\sqrt{3})^2$ was then correctly computed in part (b). Those who relied on some sort of grid method fared particularly badly.

In part (b) it was good to see that almost everyone was able to use a correct conjugate; the most common error was multiplying numerator and denominator by $3\sqrt{7} - 7$. Most candidates then went on to earn the further method mark, though a few did not combine $\sqrt{3} \times \sqrt{7}$ to make $\sqrt{21}$ and others could not simplify $\sqrt{7} \times \sqrt{7}$ etc. A large number of candidates did all the correct

working but did not simplify $\frac{15 + 5\sqrt{21}}{20}$ to earn the final mark.

Q14.

There was a wide range of responses to this question and relatively few candidates gained full marks.

In part (a)(i) most candidates wrote down a correct form for $\tan 2x$ with only a few writing $\tan^2 x$ instead of $2\tan x$. Most multiplied by the denominator to obtain a cubic equation in $\tan x$. However, many divided throughout by $\tan x$ to get $\tan^2 x = 3$ and failed to demonstrate that $\tan x = 0$ is also a solution. Some used verification which did not warrant any marks.

In (a)(ii) very few candidates gained this mark. The solution of 120° was almost always omitted. Most wrote 0° and 180° too. This in itself was not penalised this time as these

values were outside the range of x requested.

Part (b)(i) was answered very well, as candidates showed familiarity with double angle formulae and handled the algebra involved with confidence.

In part (b)(ii) the quadratic formula was used correctly by most candidates and most demonstrated where the $\sqrt{3}$ came from. The majority of candidates, though, just seemed to ignore the negative root and gave no explanation as to why. If they found only one root then they lost 2 marks. Some gave unsatisfactory explanations for the exclusion of the negative root, such as $\sin x$ cannot be negative or that it was > 1 . Those who used completion of the square to solve the equation almost invariably just took the positive square root.

Q15.

In part (a)(i) the term $\sqrt{48}$ was usually expressed as $4\sqrt{3}$ although many left their answer as $2\sqrt{12}$.

In (a)(ii) those who multiplied by $\frac{\sqrt{12}}{\sqrt{12}}$ made little progress. The most successful used the hint

from part (i) and expressed each term in the form $k\sqrt{3}$; many who obtained $\frac{10\sqrt{3}}{2\sqrt{3}}$ believed that

this simplified to $5\sqrt{3}$. Consequently, the final correct answer was only obtained by the better candidates.

In part (b) it was pleasing to see that most candidates were familiar with the technique for rationalising the denominator and, although there were some who made slips when multiplying out the two brackets in the numerator, most obtained the correct answer in the given form. Once again, some candidates cancelled incorrectly and thus did not obtain the correct final answer.

Q16.

In part (a), there were far more mistakes than had been anticipated; for example,

$\sqrt{50} = 2\sqrt{5}$ and $\sqrt{18} = 2\sqrt{3}$. It was also common to see poor cancelling such as

$$\frac{8\sqrt{2}}{2\sqrt{2}} = 4\sqrt{2}$$

. Examiners had to take care that totally incorrect work leading to the correct

$$\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}} = \frac{2\sqrt{25} + 2\sqrt{9}}{2\sqrt{4}} = \frac{10 + 6}{4} = 4$$

answer was not rewarded. For instance, $\frac{\sqrt{50} + \sqrt{18}}{\sqrt{8}} = \frac{2\sqrt{25} + 2\sqrt{9}}{2\sqrt{4}} = \frac{10 + 6}{4} = 4$ was seen on a number of occasions.

In part (b), it was pleasing to see that most candidates were familiar with the technique for rationalising the denominator in this type of problem and, although there were some who made slips when multiplying out the two brackets in the numerator, particularly when

trying to calculate $2\sqrt{7} \times 2\sqrt{7}$, many obtained the correct answer in the given form and it was good to see most getting the final step correct by dividing **both** terms by 3.

Q17.

Part (a) Many candidates were successful with this part, although sign errors and arithmetic slips were common.

Part (b) Most candidates recognised the first crucial step of multiplying the numerator and

denominator by $1 - \sqrt{5}$ and many obtained $\frac{44 - 20\sqrt{5}}{-4}$, but then inaccurate evaluation of the numerator or poor cancellation led to many failing to obtain the correct final answer.

Q18.

In part (a) it was pleasing to see that most candidates were familiar with the technique for rationalising the denominator in this type of problem and, although there were some who made slips when multiplying out the two brackets in the numerator, most obtained the correct answer in the given form.

In part (b) the term $\sqrt{45}$ was usually expressed as $3\sqrt{5}$, but the term $\frac{20}{\sqrt{5}}$ caused far more difficulties than expected. Consequently, the final correct answer was only obtained by the better candidates.

Q19.

In part (a) most candidates recognised the first crucial step of multiplying the numerator

and denominator by $3 + \sqrt{7}$ and many obtained $\frac{22 + 8\sqrt{7}}{2}$, but then poor cancellation led to a very common incorrect answer of $11 + 8\sqrt{7}$.

Candidates found part (b) more difficult than part (a) and revealed a lack of understanding of surds. Most candidates realised the need to use Pythagoras' Theorem but many could

not square $2\sqrt{5}$ and $3\sqrt{2}$ correctly. Little credit was given for those who wrote things such as $x = \sqrt{20} - \sqrt{18} = \sqrt{2}$ and candidates need to realise that "getting the right answer" is not always rewarded with full marks. Although the equation $x^2 = 2$ has the solution $x = \pm \sqrt{2}$, it was necessary to consider the context and to give the value of x as $\sqrt{2}$.

Q20.

Candidates did not always approach part (a) of the question with confidence. Several

wrote $5\sqrt{8} = 5\sqrt{4 \times 2} = 7\sqrt{2}$ or $5 + 2\sqrt{2}$; others tried to rationalise $\frac{6}{\sqrt{2}}$ by simply multiplying the denominator by $\sqrt{2}$. Consequently, it was quite common to see only one

of the two terms expressed correctly in the form $k\sqrt{2}$. It was quite strange, though, to see many obtaining an answer of $13\sqrt{2}$ from completely wrong working; clearly this was not given any credit. Some combined the two terms with a common denominator but often with an incorrect numerator.

In part (b), it was not uncommon to see the denominator and numerator multiplied by different surds and the usual errors occurred as candidates tried to multiply out brackets. A few multiplied top and bottom by the conjugate of the numerator. Nevertheless, this part of the question seemed to be answered much better than similar questions in previous years, despite the fairly difficult denominator.

Q21.

Far too many candidates believe that $\sqrt{4} = \pm 2$. Clearly some candidates had not covered this topic. However, a good proportion of the entry scored full marks.

In part (a), many left their answer as $\sqrt{36}$ or $3\sqrt{4}$, while there were many incorrect answers such as $4\sqrt{3}$.

In part (b), many earned the method mark for, say, multiplying by $\frac{\sqrt{3}}{\sqrt{3}}$ or converting $\sqrt{12}$ to $2\sqrt{3}$. However, not all could correctly complete their solution. The most common error was writing $\frac{2\sqrt{3}}{\sqrt{3}} = \sqrt{3}$.

There were two approaches for part (c). Some substituted for x and y and then attempted to multiply out the two brackets. Most simplified $(\sqrt{3})^2$ and $(\sqrt{12})^2$ successfully, but the simplification of $2\sqrt{3}\sqrt{12}$ caused problems. The other popular approach was to add $\sqrt{3}$ and $2\sqrt{3}$ and then square $3\sqrt{3}$. Unfortunately the brackets were often omitted and a final answer of 9 was presented.

Q22.

Part (a) It was not uncommon to see the denominator and numerator multiplied by different surds and the usual errors occurred as candidates tried to multiply out brackets. This part of the question did not seem to be answered as well as in previous years.

Part (b)(i) was usually correct but very few were successful in solving the equation in part (b)(ii), even though they reached forms of the correct equation such as $2\sqrt{5x} = 4\sqrt{5}$ or

$$x = \frac{4\sqrt{5}}{2\sqrt{5}}.$$

Q23.

Part (a) Some candidates found this part more difficult than part (b) and revealed a lack of understanding of surds. Some managed to express the first term as $\sqrt{7}$ but were unable to deal with the second term. Those who attempted to find a common denominator often multiplied the terms in the numerator and/or added those in the denominator. Very few obtained the correct answer of $3\sqrt{7}$.

Part (b) Most candidates recognised the first crucial step of multiplying the numerator and denominator by $\sqrt{7} + 2$ and many obtained $\frac{9 + 3\sqrt{7}}{3}$, but poor cancellation led to a very common incorrect answer of $3\sqrt{7} + 3$.

Q24.

Many candidates earned full marks on this introductory question.

- (a) Most candidates multiplied out the two brackets to obtain four terms. The most common error occurred in the last term, which was sometimes seen as -2 instead of -4 . Very few candidates recognised that it was the difference of two squares.
- (b) This part was less well done. Some candidates had problems simplifying $\sqrt{8}$ and $\sqrt{18}$ and wrote $2\sqrt{4}$ and $2\sqrt{9}$, for example. Some, having correctly converted both surds, added them incorrectly and so $6\sqrt{6}$ was quite common. A few candidates thought $\sqrt{8} + \sqrt{18}$ were equal to $\sqrt{26}$.

Q25.

Part (a) Almost everyone recognised that multiplication of the two brackets was required but there were numerous errors with $7\sqrt{5}$ instead of $12\sqrt{5}$ being common and -2 or -4 instead of -3 . Although most dealt with the first term correctly and obtained 20, many added $12\sqrt{5}$ and $-\sqrt{5}$ wrongly to get $-11\sqrt{5}$.

Part (b) This part was answered more successfully with, $\sqrt{\frac{75}{3}} - \sqrt{\frac{27}{3}}$ being the neatest

method. Some 2-73 failed to complete correctly from $\frac{2\sqrt{3}}{\sqrt{3}}$ to 2 and gave an answer of $\sqrt{3}$. A few went 'all round the houses' but got there eventually. Some tried to cancel out $\sqrt{3}$ but only considered one term in the denominator. Multiplying top and bottom by $\sqrt{3}$ caused some problems. A few attempted to combine the 2 terms in the numerator

and wrote $\frac{\sqrt{48}}{\sqrt{3}}$ which of course is also an integer!

Q26.

The difference of two squares was often not recognised in part (a). Nevertheless, most were able to multiply out the brackets and, those who realised that $(\sqrt{12})^2$ was equal to

12, usually scored full marks for part (a).

Practically everyone knew that $\sqrt{12} = 2\sqrt{3}$ in part (b).

However, the vast majority of candidates failed to express their final answer to part (c) in the required form. Even the most able candidates were content to leave their answer as $2 + \frac{1}{2}\sqrt{12}$.

Q27.

This was a straightforward question on coordinate geometry intended to get the candidates off to a confident start and many scored full marks. However many made very fundamental errors.

Part (a) Instead of finding the arithmetic mean of the pair of x -coordinates and the pair of y -coordinates, many candidates found the difference of the coordinates before dividing by 2. A few were able to deduce the coordinates of the mid-point from a diagram and some used the half line vector approach successfully.

Part (b) There were many correct answers but a small number were incorrect and some candidates wrote $\sqrt{52} = 4\sqrt{13}$. Notational errors such as $36 + 16 = \sqrt{52}$ were seen and will not always be condoned. It was surprising how many candidates made no attempt at the distance formula.

Part (c)(i) Although the majority of answers were correct, an alarming number of candidates could not simplify $-6/-4$ correctly. It was also very disappointing to see many using difference in x values/difference in y values in order to find the gradient of a line.

Part (c)(ii) Candidates were expected to use their gradient to determine the equation of the line and many had a complete proof. It was not expected that so many would use the equation $y = mx + c$ at this stage. Many fudged their work to get the printed answer. Some imagined that it was sufficient simply to show the required line had the same gradient as they had evaluated in part (c)(i) or to show that **one** of the points satisfied the given equation. Neither of these on their own scored any marks. Verifying two of the facts was given some credit.

Part (d) Although the majority of candidates found the point of intersection, there were many algebraic and sign errors in the attempts seen – adding equations after equalising coefficients being the most common. Many students did not recognise what was required and made no attempt at it.

Q28.

Part (a) The most common error was to multiply $\sqrt{3}$ by $\sqrt{3}$ to get 9 instead of 3.

Part (b) Of those candidates who multiplied both numerator and denominator by the correct expression, only about half simplified correctly and $2 + 2\sqrt{3}$ was a common incorrect answer. Many of the rest multiplied by $\sqrt{3} - 1$ instead, which led nowhere.

Q29.

Many candidates were able to give the correct expansion in part (a), normally by applying

Pascal's triangle. Many realised that x should be replaced by reaching the exact printed answer. Candidates should be aware that approximate decimal values for each of 4

$\sqrt{5}$ and $4(\sqrt{5})^3$ cannot then be added and the result used to equate to the exact surd part of the printed answer. The final part of the question proved difficult, although, again, some elegant solutions were presented. Those who wrote ' $\log_2(1 + \sqrt{5})^4 = \log_2(56 + 24\sqrt{5}) = \log_2[8(7 + 3\sqrt{5})]$ ' usually went on to score all three marks.



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